INCH-POUND

MIL-DTL-83361B
27 November 2007
SUPERSEDING
MIL-DTL-83361A
23 February 1996

DETAIL SPECIFICATION

INDICATOR, PRESSURE, OIL, DUAL, 0-100 PSI, EGU-18/A

Inactive for new design after 1 May 1995.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE. This specification covers one type of hermetically sealed, remote indicating, synchro-style, integrally lighted, dual oil-pressure indicator, designated EGU-18/A.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-595	- Colors Used in Government Procurement (Color chips 17038, 37038, and 37875).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MS3126	- Connectors, Plug, Electrical, Crimp Type, Straight, Bayonet Coupling.
MIL-DTL-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-S-7742	- Screw Threads, Standard, Optimum Selected Series: General Specification for.
MIL-A-8625	- Anodic Coatings for Aluminum and Aluminum Alloys.
MIL-C-14806	- Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges.
MIL-T-25624	- Transmitter, Pressure TRU-80/A, Oil, Engine-Mounted, 0-100 PSI.
MIL-L-27160	- Lighting, Instrument, Integral, White General Specification for.
MS28005	- Transmitters, Pressure, Synchro, Aircraft.
MS33558	- Numerals and Letters, Aircraft Instrument Dial, Standard Form of.
MS33639	- Cases, Instrument, Clamp-Mounted, Aircraft.
MS33678	- Connector, Receptacle, Electrical, Integral Mounting.
MIL-DTL-83488	- Coating, Aluminum, High Purity.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-202	- Electronic and Electrical Component Parts.
MIL-STD-461	- Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment.
MIL-STD-464	- Electromagnetic Environmental Effects Requirements for Systems.
MIL-STD-704	- Aircraft Electric Power Characteristics.
MIL-STD-810	- Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-889	- Dissimilar Metals.

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454 - General Guidelines for Electronic Equipment.

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM28042 - Clamp, Instrument Mounting, Aircraft.

(Copies of this document are available online at http://www.aia-aerospace.org/ or from Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22009.)

ASME INTERNATIONAL

ASME Y14.34M - Associated Lists.

(Copies of this document are available online at http://www.asme.org/ or from ASME International, Three Park Avenue, New York, NY 10016-5990.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

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3.2.3 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator, except where ferrous materials are essential.

3.2.4 Metals. Metals shall be corrosion resistant or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be encountered in storage or during normal service.

3.2.4.1 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-DTL-83488, type I or II, as applicable, and of a class that is adequate to achieve the degree of protection required. Other protective coating in lieu of MIL-DTL-83488 may be used if demonstrated to be satisfactory and is approved by the procuring activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.2.4.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other.

3.2.5 Toxic and corrosive fumes. The indicator shall be of such material and construction that when burned, fumes or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.2.6 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack , chip, or scale with age or extremes of climate and environmental conditions shall be avoided.

3.3 Design and construction. The indicator shall contain two synchro receivers and shall be designed for use with the TRU-80/A transmitter conforming to MIL-T-25624 and MS28005. The indicator shall be designed so that parallel connection of corresponding synchro receivers of two indicators with a single transmitter will not degrade the performance of the indicators nor the transmitter.

3.3.1 Case. The case shall be hermetically sealed. The dimensions shall conform to MS33639 and figure 1.

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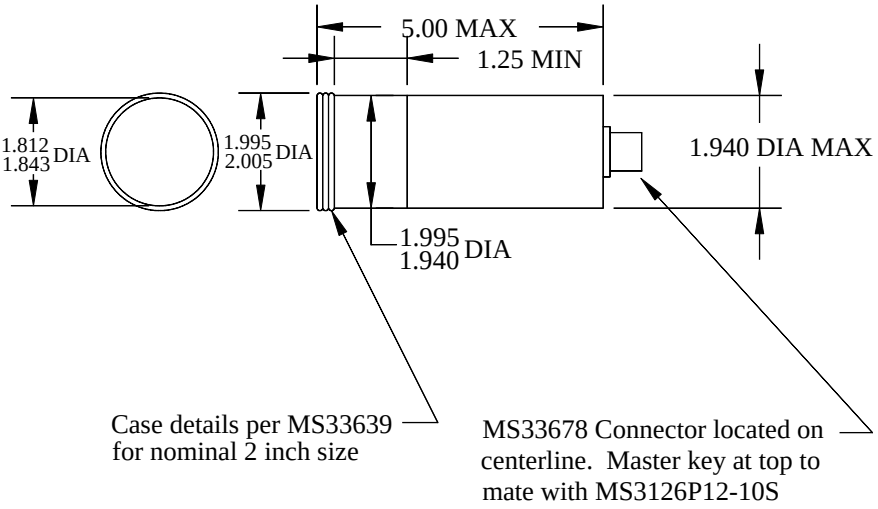


FIGURE 1. Outline dimensions.

3.3.1.1 Cap. A separate nonhermetically sealed cap may be used on the back of the case for installing the electrical receptacle. The cap shall be made of nonferrous, low-density metal.

3.3.1.2 Case-to-coverglass distance. The distance between the outer surface of the coverglass and the front edge of the case shall be the minimum practicable and shall not exceed 0.031 inch.

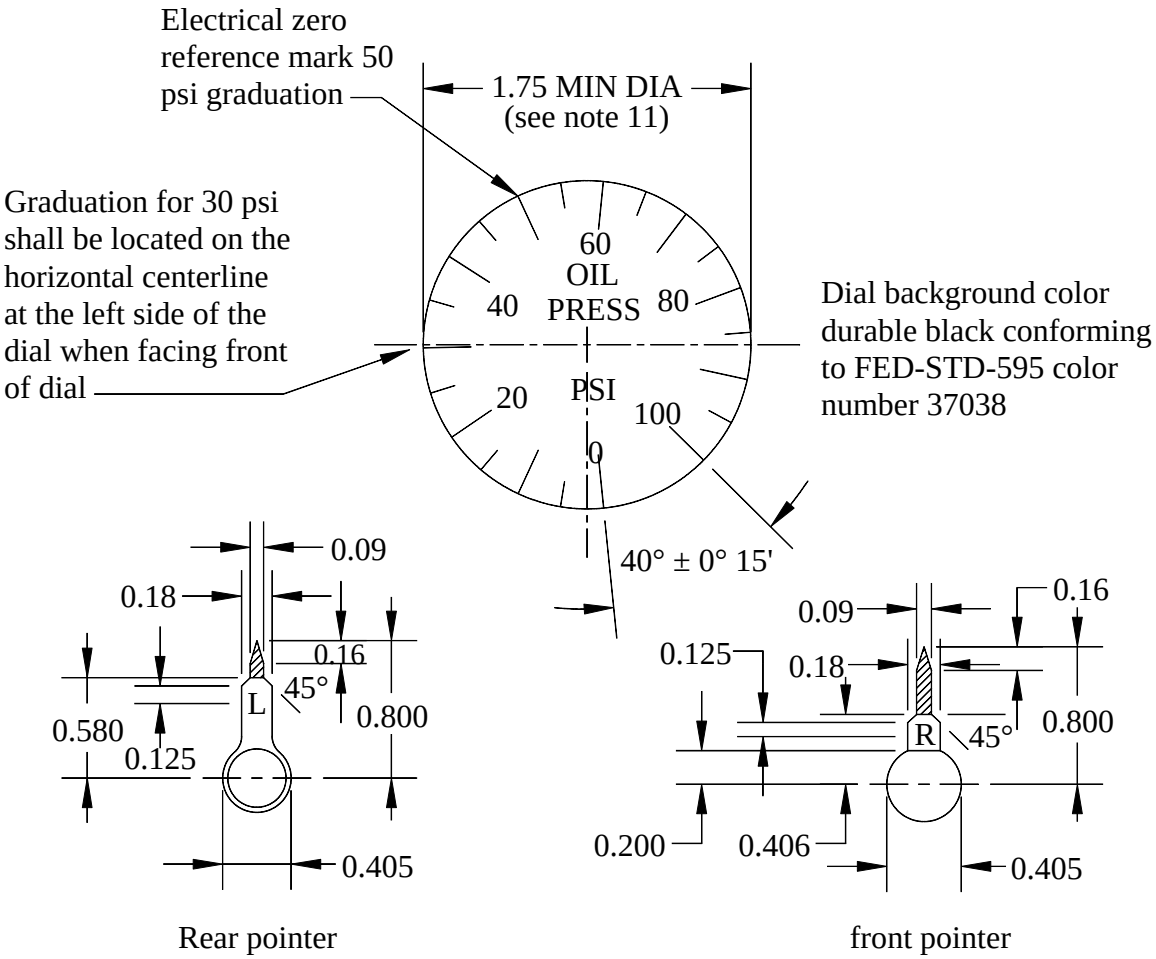
3.3.1.3 Filling medium. The filling medium shall be at least 98 percent pure and shall be free of dust particles. The filling medium shall contain no more than 0.006 milligram of water per liter (-85 °F (-65 ºC) dew point) at the filling pressure. The filling medium shall be 88 percent to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 pounds per square inch (psi) to 14.7 psi. Other concentrations of filling medium shall not be used unless prior approval from the procuring activity is obtained.

3.3.2 Lighting. The indicator shall be integrally lighted in accordance with MIL-L-27160 and shall meet all requirements specified herein.

3.3.2.1 Prime standard. A prime standard shall be provided by the supplier which shall be either an indicator conforming to this specification or a lighted model. The prime standard lighting shall conform to MIL-L-27160. The design, color, quality of lighting, and calibration shall be subject to the approval of the procuring activity.

3.3.3 Pointers. Pointers shall conform to figure 2 and shall be suitable for integral lighting. The thickness of the pointers shall not be less than 0.01 inch.

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NOTES:
1. Shaded pointer areas, pointer letters, and dial markings are to be lighted.
2. Dimensions in inches.
3. Unless otherwise specified, tolerances are: 2 Places ±0.10; 3 places ±0.005, angles +1 degree.
4. Color of dial markings specified in 3.3.4.1: durable dull white conforming to FED-STD-595 color number 37875.
5. Color of dial markings specified in 3.3.4.2: durable dull black conforming to FED-STD-595 color number 37038, durable gloss black conforming to FED-STD-595 color number 17038, or durable dull white conforming to FED-STD-595 color number 37875.
6. For dial markings dimensions, see table I of this specification.
7. Each graduation shall be within ±0º 15' of the true linear scale position.
8. Dial background and unshaded part of pointers shall be durable dull black conforming to FED-STD-595 color number 37038.
9. Thickness of pointers shall not be less than 0.01 inch.
10. Numerals and letters shall be in accordance with MS33558.
11. Outside diameter shall be such that the pointer shall overlap one-fifth of the shortest graduation on the dial.

FIGURE 2. Dial and pointers.

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3.3.3.1 Pointer mechanism. The indicator shall be provided with two pointers that shall be attached to the concentric shafts so that the rear pointer will be actuated from the front synchro and the front pointer will be actuated from the rear synchro. The direction of rotation of the synchro shafts, as viewed from the end at which the pointers are attached, shall be clockwise with increasing pressure.

3.3.4 Dial. The dial shall conform to figure 2. A minimum diameter of 1.75 inches shall be maintained across the outside ends of the graduations.

3.3.4.1 Dial markings. The following markings shall be finished in durable white conforming to FED-STD-595 color number 37875. The dimensions of the markings shall be as specified in table I.

TABLE I. Dimensions for dial markings.

Marking	Height or length inch ± 0.015	Width of line or graduation inch ± 0.005
Numerals 0, 20, 40, 60, 80, 100	0.125	0.016
10-pound graduations	0.200	0.025
Graduation at 50 psi	0.200	0.037
Lettering, "OIL PRESS"	0.125	0.160
Lettering, "PSI"	0.093	0.015
5-pound graduations	0.093	0.015

3.3.4.2 Other markings. The marking, "EGU-18/A", shall be permanently and legibly marked on the dial in letters 0.062 inch high. Position of this marking on dial face is at the discretion of the manufacturer. This marking and the color of all other markings not otherwise specified shall be finished in durable dull black conforming to FED-STD-595 color number 37038, durable gloss black conforming to FED-STD-595 color number 17038, or durable dull white conforming to FED-STD-595 color number 37875.

3.3.4.3 Dial visibility. Portions of the minor graduations of the dial shall be visible when viewed from any position within the frustrum of a cone, which is the side that intersects the outer edge of the coverglass and makes an angle of 30 degrees with the perpendicular through the center of the dial.

3.3.4.4 Dial-to-coverglass distance. The distance between the indicator dial and the inside of the coverglass shall be the minimum practicable and shall not exceed 0.188 inch.

3.3.5 External bonding. Provisions shall be made for electrically bonding the indicator case to the aircraft structure. Bonding shall be in accordance with MIL-STD-464.

3.3.6 Coverglass. The coverglass shall be clear, flat, and free of flaws that would interfere with the normal reading of the instrument. The thickness of the coverglass shall be the minimum practicable and entirely suitable for the purpose. The coverglass shall be provided with a reflection-reducing coating conforming to MIL-C-14806.

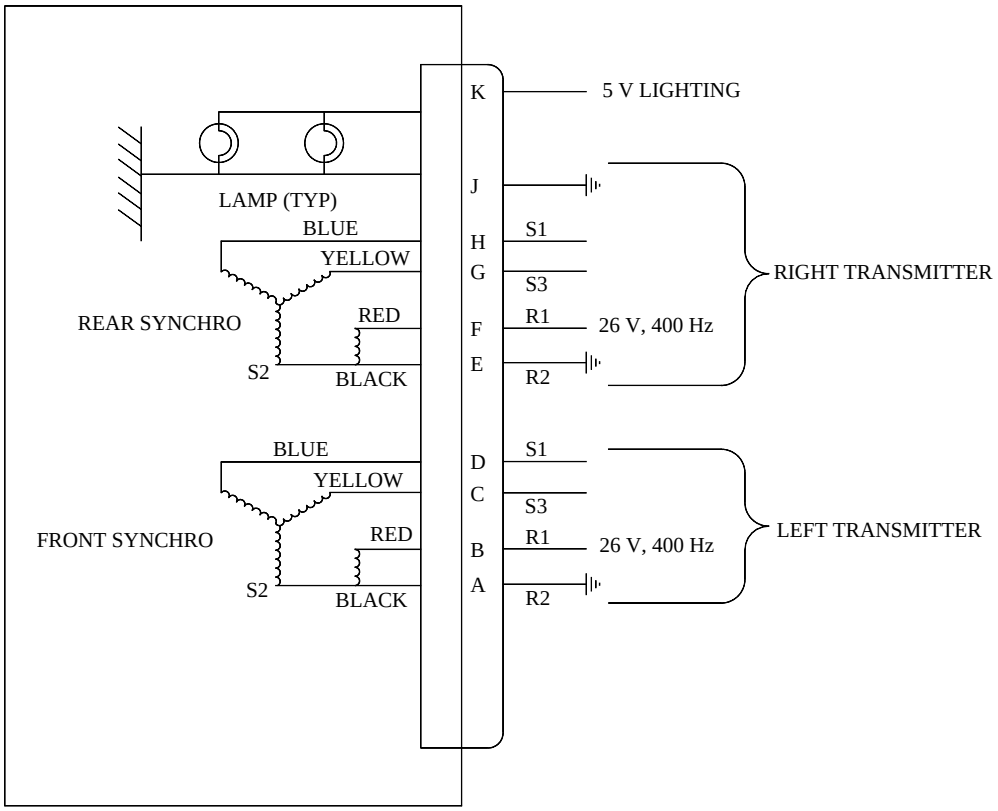
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3.3.7 Fasteners. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be located in a position where they do not interfere with the dial markings or prevent the markings from being located in the proper position.

3.3.8 Electrical zero reference mark. The dial shall be provided with an electrical zero reference mark on the scale portion equidistant in angular degrees between the first and last graduations.

3.3.9 Electrical connector. The indicator shall be provided with an electrical connector in accordance with figure 1. When the connector is nonhermetically sealed, a hermetically sealed baffle plate shall be used to seal the indicator.

3.3.10 Electrical wiring. The wiring of the indicator shall be in accordance with figure 3. The indicator wiring shall be designed so that no portion of the internal circuitry will be connected to the instrument case except the ground wire of the lighting circuit.



NOTE: Pin J of the lighting circuit shall be internally connected to the instrument case.

FIGURE 3. Wiring.

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3.3.11 Synchros. The indicator synchros shall consist of a salient two-pole wound rotor used as the primary and shall be designed for 26 volt (V) 400 Hertz (Hz) single-phase power in accordance with MIL-STD-704, with y-connected stators as secondaries. The stators shall be wound so that the amplitudes of the induced voltages measured between any two stator leads will vary approximately sinusoidally with the angular position of the rotor. Pivots, bearings, and gears shall not bind nor shake and shall be as nearly frictionless as possible.

3.3.11.1 Synchro receivers. Synchro receivers shall have electrical characteristics compatible with those defined in MIL-STD-461.

3.3.12 Secondary voltage. The maximum open-circuit voltage measured between any two secondary leads shall be 11.8 V ± 3 percent.

3.3.13 Reliability (life). The indicator shall have a mean-time-between-failure (MTBF) of 2,000 hours at a 50 percent confidence level.

3.3.14 Maintainability. No preventive maintenance shall be required throughout the service life of the indicator. The mean-time-between-maintenance (MTBM) shall not exceed the MTBF. Fault isolation and repair within the hermetically sealed indicator shall be accomplished at the depot level.

3.4 Performance. The indicator shall operate as specified herein during and after subjection to the following conditions:

a. Temperatures: temperatures ranging from -80 °F to 160 °F (-62 ºC to 71 ºC).
b Vibration: vibration of 2 g from 5 Hz to 500 Hz.
c. Temperature-altitude: pressure altitude of 100,000 feet at -67 °F (-55 ºC).
d. Salt-fog: exposure to salt sea atmosphere for 48 hours without evidence of corrosion or deterioration.
e. Shock: up to 20 g for 11 milliseconds.

3.4.1 Scale error. Scale error shall not exceed the tolerances specified in tables II and III.

TABLE II. Scale error.

PSI	Tolerance ±PSI
0	1.0
20	1.0
40	1.0
60	1.0
80	1.0
100	1.0

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TABLE III. Test tolerances.

Test		Tolerance ±PSI
Position error at midscale		1.0
Friction	Room temperature	1.0
	High temperature	1.0
	Low temperature	1.0
Vibration	Pointer oscillation	1.0
	Pointer variation	1.0
Scale error	Low temperature	1.0
	High temperature	1.0

3.4.2 Friction error. Friction error shall not exceed the tolerances specified in table III.

3.4.3 Position error. Position error shall not exceed the tolerances specified in table III.

3.4.4 Dielectric strength. The dielectric strength between each pin of the electrical receptacle, except pins J and K and the case and between all external connections of each mutually insulated circuit, shall be a minimum of 500 V alternating current (AC).

3.4.5 Case leakage. Case leakage shall not exceed 0.1 micron cubic foot per hour.

3.4.6 Fogging. There shall be no evidence of moisture or oil fogging of the coverglass when the indicator is subjected to a temperature of 185 °F ± 9 °F (85 ºC ± 5 ºC) for a minimum period of 2 hours and the face of the coverglass is then immediately immersed in an ice bath for 3 minutes.

3.4.7 Effect of lead length. When the lead length between the indicator and transmitter is changed from 5 to 500 feet, the indicator shall not change more than 1 degree of arc. The ground lead shall be kept short.

3.4.8 Magnetic effect. The compass deflection shall not exceed 1 degree of arc when the indicator is positioned 5.5 inches from the compass and the horizontal magnetic field intensity at the indicator is 0.17 oersted (Oe) to 0.19 Oe.

3.4.9 Unit torque gradient. The unit torque gradient shall be a minimum of 2,000 milligram-millimeters (mg-mm) per angular degree of rotation.

3.4.10 Voltage and frequency variation. The indicator shall withstand voltage and frequency variations from 22 V to 30 V and 320 Hz to 480 Hz without exceeding the scale error and friction error tolerances specified in tables II and III.

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3.4.11 Insulation resistance. The insulation resistance between each pin of the electrical receptacle and the case and between all external connections of each mutually insulated circuit shall be a minimum of 100 megohms. This requirement does not apply to the lighting circuit (pins J and K).

3.4.12. Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 for class A1 equipment. The test methods shall be as specified in 4.5.19.

3.5 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of ASME Y14.34M shall govern the manufacturer's part numbers and changes thereto.

3.6 Numeral and letters. Numerals and letters shall conform to MS33558.

3.7 Finishes and protective coatings. Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625. Small holes, straight threads, case inserts, and parts in hermetically sealed enclosures need not be anodized. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-DTL-5541. When abrasion resistance is a factor, chemical films in accordance with MIL-DTL-5541 shall not be used.

3.8 Weight. The weight of the indicator shall not exceed 1.0 pound.

3.9 Screw threads. Unless otherwise specified herein, the threads of all machine screws shall conform to MIL-S-7742.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10.1 Nameplate. A nameplate shall be permanently attached to the rear of the case. Where applicable, the information marked thereon shall conform to MIL-STD-130. The nameplate shall not be glued or cemented to the case nor shall metal decals be used.

3.11 Pointer or movement interference. In tests that require operation of the indicator, the inspection methods are intended for each movement of the indicator. Tests of the movements may be conducted individually or simultaneously, unless otherwise specified. A check of each movement through its entire range to ensure that there is no interference with the other movement or any other type of interference shall be conducted in each operational test.

3.12 Workmanship. The workmanship guideline of MIL-HDBK-454 may be used as a reference for the determination of quality workmanship.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions.

4.2.1 Standard atmospheric conditions. Unless otherwise specified herein, all required tests shall be performed under the following conditions:

a. Temperature - ambient room (approximately 75 °F (25 °C)).
b. Pressure - normal atmospheric (approximately 29.92 inches of mercury (Hg)).

When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.2.2 Attitude. Unless otherwise specified herein, the indicator shall be tested in the normal operating position with the plane of the dial vertical.

4.2.3 Electrical power. Unless otherwise specified herein, the indicator shall be tested with 26 V ± 1 V, single-phase, 400 Hz ± 5 Hz, AC power and with 4.5 V ± 0.5 V AC or DC applied to the lighting circuit.

4.2.4 Tapping. Unless otherwise specified herein, the indicator shall be lightly tapped or vibrated before each reading.

4.2.5 Integral lighting. The integral lighting tests shall be conducted in accordance with MIL-L-27160 in the sequence specified.

4.2.6 Test tolerances. The indicator shall operate within the tolerances specified in tables II and III.

4.3. First article inspection. The first article inspection shall consist of all the tests in the order specified in paragraph 4.5.

4.3.1 First article inspection sample. The first article test samples shall consist of two indicators of the same design and construction that shall have the same performance and physical characteristics as production indicators.

4.3.2 Nonstandard parts data approval. When nonstandard electronic parts are used in the construction of the indicator, nonstandard parts data approval shall be a prerequisite to first article approval.

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4.4 Conformance inspection. Conformance inspection shall consist of the following:

a. Individual tests (see 4.4.1).
b. Sampling tests (see 4.4.2).

4.4.1 Individual tests. Each indicator shall be subjected to the following tests:

a. Examination of product (see 4.5.1).
b. Individual tests of MIL-L-27160 and visual comparison (see 4.5.2 and 4.5.2.1).
c. Dielectric strength (see 4.5.3).
d. Electrical zero (see 4.5.4).
e. Scale and friction error (see 4.5.5).
f. Position error (see 4.5.6).
g. Sealing (see 4.5.7).

4.4.2 Sampling tests. One indicator shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests:

a. Individual tests (see 4.4.1).
b. Sampling plan A tests of MIL-L-27160 (see 4.5.8).
c. Vibration (see 4.5.9).
d. Low temperature (see 4.5.10).
e. High temperature (see 4.5.11).
f. Fogging (see 4.5.12).
g. Insulation resistance (see 4.5.13).

4.5 Tests.

4.5.1 Examination of product. The indicator shall be examined to determine compliance with the requirements specified herein with respect to the following:

a. Exterior finish.
b. Case and outline dimensions.
c. Display in accordance with 3.3.3 and 3.3.4 (includes verification of readability and visibility).
d. Electrical connector.
e. External bonding.
f. Coverglass and reflection reducing coating.
g. Weight.
h. Nameplate and markings.
i. Workmanship.

4.5.2 Lighting (individual tests of MIL-L-27160). The indicator shall be subjected to the individual tests of MIL-L-27160.

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4.5.2.1 Visual comparison. The prime standard and the indicator under test shall be placed side-by-side at eye level. The two indicators shall be viewed from a distance of 2 feet within a sighting cone of 30 degrees from normal to the surface of the coverglass. The prime standard lighting circuit shall be energized with 2.50 V ± 0.01 V and the lighting circuit of the indicator under test shall be energized with 2.60 V ± 0.01 V. The overall brightness level of the dial and pointer of the indicator under test shall equal or exceed the brightness of equivalent areas of the prime standard. The voltage applied to the indicator under test shall be decreased to 2.40 V ± 0.01 V. The brightness levels of the dial and pointer of the prime standard shall be equal to or less than the brightness of the equivalent areas of the indicator under test. A voltage of 4.50 V ± 0.01 V shall be applied to both the prime standard and the indicator under test. The two units shall be compared for white color and shall appear the same. All indicators that fail this test shall have the questionable areas measured with a photometer. If they also fail the photometer test, the indicators shall be rejected.

4.5.3 Dielectric strength. The dielectric strength test shall be conducted in accordance with MIL-STD-202, Method 301, "Dielectric Withstanding Voltage", after the indicator has reached its normal operating temperature. A voltage of 500 V AC at commercial frequency shall be applied between the external connections of each mutually insulated circuit and between each external connection and the case. Any arc-over, breakdown, or current flow that constitutes a transient or steady increase in leakage current at the specified test voltage shall be cause for rejection. This test shall not include the lighting circuit.

4.5.4 Electrical zero. A voltage of 26 V ± 1 V AC single-phase power shall be applied to the rotor leads designated as R1 and R2 (see figure 3). R2 shall be connected to the ground or low potential side of the 26 V AC power supply. With the S2 lead connected to the R2 lead, the following conditions shall exist at electrical zero (50 psi):

a. The S3S1 voltage shall be minimum, shall be in-phase with voltage R2R1, and shall remain approximately in-phase as the synchro rotor is rotated from a position corresponding to 50 psi, decreasing to a position corresponding to -6 psi.
b. The R1S3 voltage shall be near minimum at a rotor position corresponding to 50 psi and shall reach minimum when the synchro rotor is rotated to a position corresponding to 59.375 psi.
c. Rotation of the synchro rotor from a position corresponding to 50 psi increasing to 59.375 psi shall produce an increasing S3S2 voltage and a decreasing S2S1 voltage.

4.5.5 Scale and friction error. The indicator shall be tested for scale error at the test points specified in table II. The indicator shall be lightly tapped or vibrated before each reading to remove normal instrument friction. The scale errors shall not exceed the tolerances specified in table II. The friction error test may be conducted in conjunction with the scale error test. The difference between the readings before and after the indicator is tapped or vibrated shall constitute the friction error. The friction error shall not exceed the tolerances specified in table III. This procedure shall be accomplished both up-scale and down-scale at the test points specified.

4.5.6 Position error. The indicator pointer shall be positioned at electrical zero reference mark with power applied. The change in pointer position when the indicator is held in any position shall not exceed the value specified in table III.

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4.5.7 Sealing. The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leakage rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.5.8 Lighting. The indicator shall be subjected to the sampling plan A tests of MIL-L-27160.

4.5.9 Vibration. Prior to conducting this test, the test indicator shall be operated under standard ambient conditions and a record shall be made of all data necessary to determine compliance with the requirements of 4.5.5. The entire test shall be conducted at ambient temperature with the indicator properly connected and with power applied. The lighting circuit shall be energized with 4.5 V + 0.5 V, -0.0 V AC. The three mutually perpendicular axes of applied vibration and the location where the acceleration and excursion(s) shall be monitored are shown in figure 4. Duration of the various portions of this test shall be as specified in table IV. No more than four resonances shall be tested in each axis.

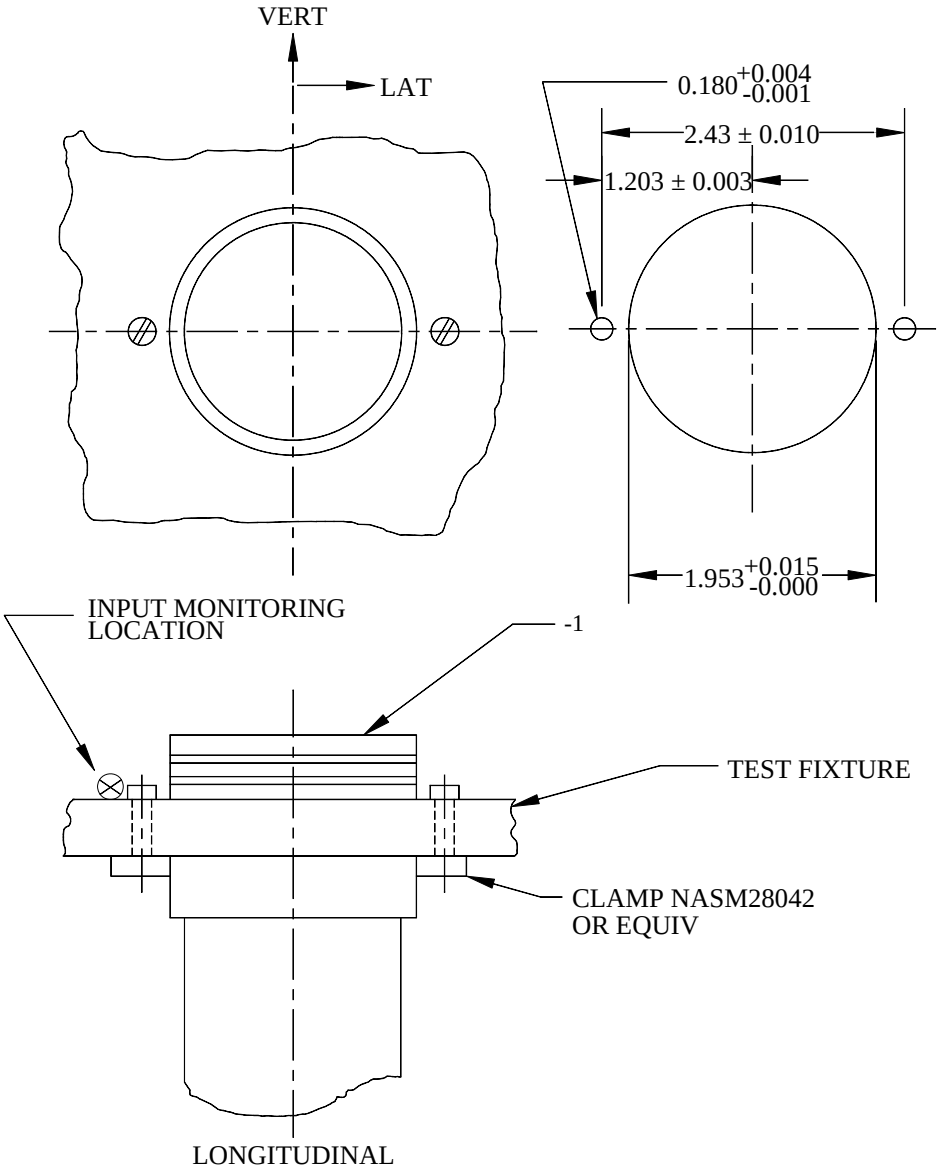


FIGURE 4. Installation for vibration test.

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TABLE IV. Sinusoidal test schedule.

Number of resonances	0	1	2	3	4
Total vibration time at resonance	0	30 minutes	1 hour	1-1/2hours	2 hours
Cycling time	3 hours	2-1/2 hours	2 hours	1-1/2 hours	1 hour

NOTES:
1. The total vibration time at each resonance shall be 30 minutes.
2. The above time durations apply to testing along one axis only; three axes are required for a complete test.

4.5.9.1 Sinusoidal cycling and resonance conditions. The order in which the cycling and resonance portion of this test are conducted shall be optional. The pointer shall be set at each of the following indications for 1 hour during vibration along each axis:

a. 20 psi.
b. 50 psi.
c. 80 psi.

NOTE: Resonant modes of the specimen shall be determined by slowly varying the frequency between the limits of 5 Hz and 500 Hz at an amplitude of applied vibration not exceeding that shown by the cycling test curve. When dynamically complex test fixtures or equipment are involved, low input vibration levels shall be employed to avoid unnecessary failures during the resonance search. The resonances selected for this test shall be those that are most likely to produce failure. If more than four resonances are encountered with vibration applied along any one axis, the four most severe resonances shall be selected for testing. Should any functional out-of tolerance conditions occur during the cycling test, one of the resonances tested shall be at the frequency at which the worst out-of-tolerance condition occurred. If a change in the resonance frequency occurs during the resonance dwell, the time of occurrence shall be noted and the frequency shall be adjusted immediately to maintain the peak resonance condition. The final resonance frequency shall be recorded.

4.5.9.2 Test procedure.

4.5.9.2.1 Cycling. The indicator shall be vibrated from 5 Hz to 500 Hz and back to 5 Hz at the rate shown in figure 5 and at the amplitude shown in figure 6, curve A.

4.5.9 2.2 Resonance. The indicator shall be vibrated in accordance with the resonant conditions and at the amplitude shown in figure 6, curve A.

4.5.9.2.3 Indicator operation. Operation of the indicator shall be monitored for compliance with table III. If an out-of-tolerance condition occurs, the indicator shall be subjected to and shall meet the requirements of 4.5.9.2.4.

4.5.9.2.4 Functional demonstration test. A functional demonstration test shall be conducted at the conclusion of testing in each axis with the indicator operating. It shall operate within the tolerances specified in table III. The frequency shall be varied from 5 Hz to 500 Hz and back to 5 Hz at the rate shown in figure 4 and at the amplitude shown in figure 6, curve B.

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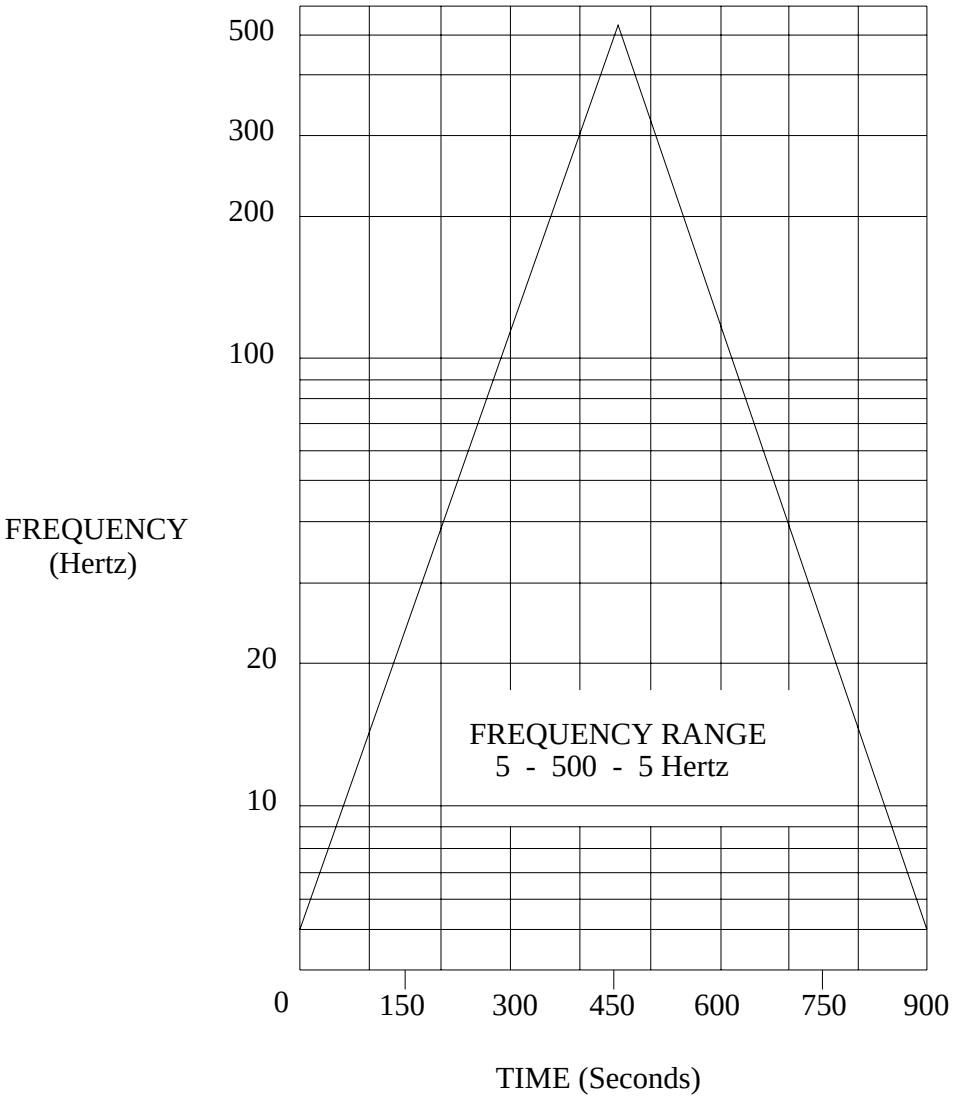


FIGURE 5. Vibration cycling test schedule.

4.5.10 Low temperature. The low temperature test shall be conducted in accordance with MIL-STD-810, Method 502, "Low Temperature", procedure for storage test, including the option of operation at low temperature and room temperature. The storage temperature shall be -80 °F (-62 ºC) and shall be held at that temperature for a period of 72 hours. At the end of the 72-hour period, the chamber temperature shall be raised to -65 °F (-54 ºC) and maintained for a 24-hour period. At the end of the 24-hour period and while still at -65 °F (-54 ºC), the indicator shall be subjected to the test specified in 4.5.5. The indicator shall not require more than 30 seconds to indicate within specified tolerances. The indicator shall then be returned to room temperature for a period of 1 hour after which it shall again meet the test specified in 4.5.5.

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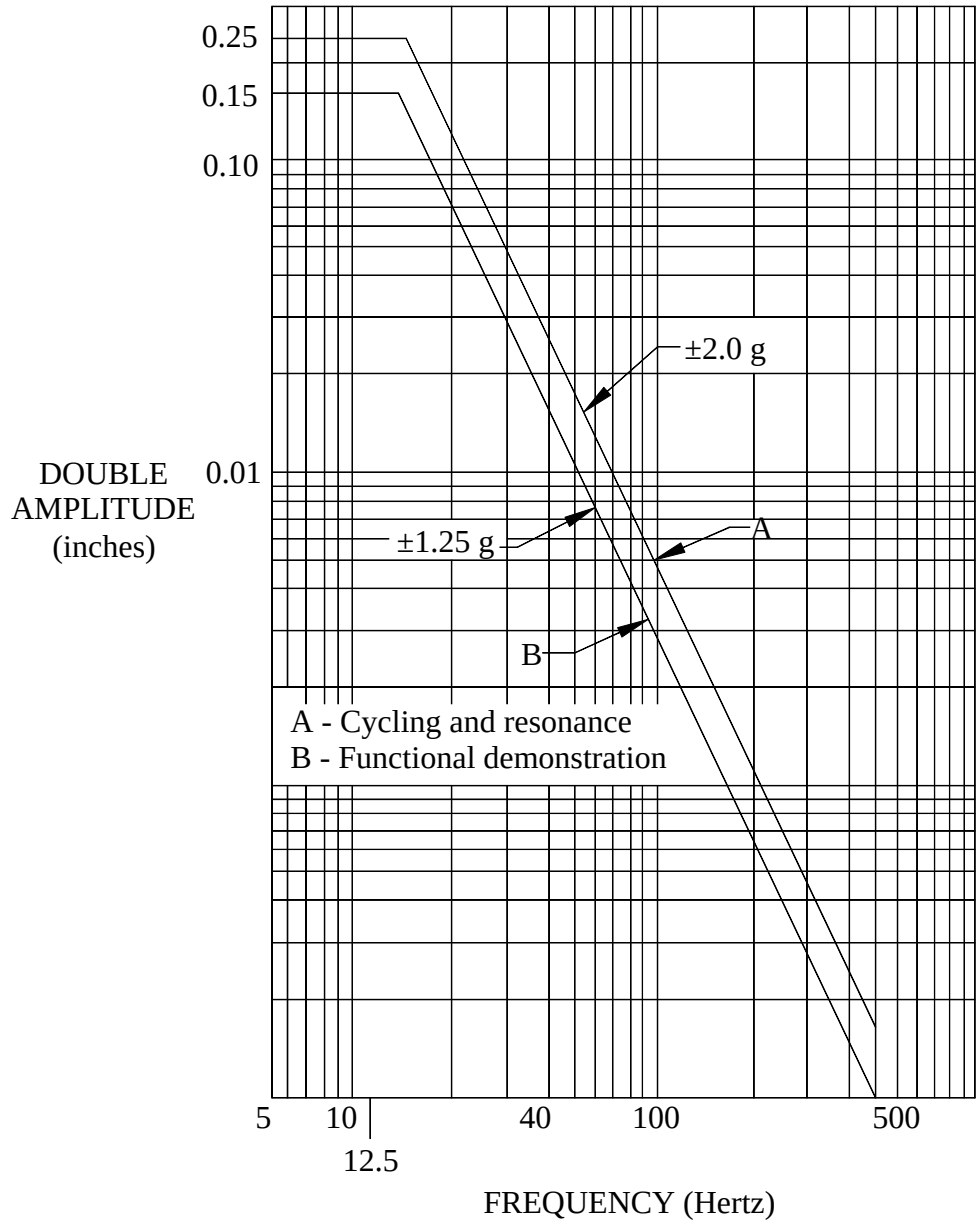


FIGURE 6. Vibration test amplitudes.

4.5.11 High temperature. The indicator shall be subjected to a high temperature test in accordance with MIL-STD-810, Method 501, "High Temperature", procedure for operational test, using the constant temperature option except the exposure time shall be at least 48 hours. The exposure temperature shall be 160 °F (71 °C). While at the high temperature, the indicator shall be subjected to and shall meet the test specified in 4.5.5. After the indicator has been returned to room temperature for 1 hour, it shall again meet the test specified in 4.5.5.

4.5.12 Fogging. The indicator shall be placed in a controlled ambient temperature of 185 °F ± 9 °F (85 ºC ± 5 ºC) for a minimum period of 2 hours. Within 10 seconds after removal from the oven, the indicator shall be immersed to a depth of 0.25 inch in an ice water bath maintained at 32 °F ± 45 °F (0 ºC to 7 ºC). After 3 minutes, the indicator shall be removed from

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the fluid bath and quickly wiped dry (compressed air shall not be used). The coverglass and dial shall immediately be inspected. Any evidence of fogging or damage shall be cause for rejection.

4.5.13 Insulation resistance. The indicator shall be subjected to an insulation resistance test in accordance with MIL-STD-202, Method 302, "Insulation Resistance", Test Condition B. The resistance shall be measured between the external connections of each mutually insulated circuit and between each external connection of the case. The insulation resistance shall be a minimum of 100 megohms. The lighting circuit shall not be included in this test.

4.5.14 Voltage and frequency variation. The indicator shall meet the scale and friction error tolerances specified in tables II and III when the voltage and frequency are varied as follows:

a. 26 V, 400 Hz.
b. 22 V, 320 Hz.
c. 22 V, 480 Hz.
d. 30 V, 320 Hz.
e. 30 V, 480 Hz.

4.5.15 Magnetic effect. The indicator, first not operating and second, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5.5 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0 degrees, 45 degrees, 90 degrees, 135 degrees, 180 degrees, 225 degrees, 270 degrees, and 315 degrees from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 Oe to 0.19 Oe. Deflection of the compass at any of the specified positions shall not exceed 1 degree.

4.5.16 Effect of lead length. The indicator shall not change more than 1 degree of arc when the lead length between the indicator and transmitter is changed from 5 feet to 500 feet of the No. 20 AWG gage wire. Equivalent resistance may be substituted for the added wire length. The ground lead shall be kept short.

4.5.17 Temperature-altitude. The temperature-altitude test shall be conducted in accordance with MIL-STD-810, Method 520, "Temperature, Humidity, Vibration, and Altitude", qualification procedure (except the temperature shall be -67 °F (-55 ºC)) with a chamber pressure equivalent to an altitude of 100,000 feet. The rate of pressure change shall not exceed an average of 3 psi per second while under these conditions. At the end of a 1-hour period, the scale and friction error test of 4.5.5 shall be conducted. The errors shall not exceed the tolerances specified in table III. This test may be conducted concurrently with the low temperature exposure test.

4.5.18 Shock. The shock test shall be conducted in accordance with MIL-STD-810, Method 516, "Shock", functional shock procedure.

4.5.19 Electromagnetic interference. The indicator shall be subjected to electromagnetic interference tests in accordance with MIL-STD-461, methods CE102, CS101, RE102, and RS103.

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4.5.20 Salt fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509, "Salt Fog", except the exposure shall be for a continuous 48-hour period. There shall be no evidence of external corrosion or deterioration that might affect subsequent operation. The scale errors after this test shall not exceed the tolerances specified for room temperature scale error.

4.5.21 Unit torque gradient. The indicator shall be connected to the master transmitter synchro and the master transmitter locked in the electrical zero position. The synchro rotors shall then be energized with rated voltage and frequency ± 1 percent. Weights shall be applied to a torque pulley that is positioned on the rotor shaft to rotate the shaft in a counter-clockwise direction through an angle of 5 degrees of arc. The torque tending to return the shaft to its original position shall be computed and shall be no less than 2,000 mg-mm per degree of displacement.

4.5.22 Lighting tests. Lighting tests shall be conducted in accordance with the sampling plan B tests of MIL-L-27160, except the dielectric test shall not be conducted.

4.5.23 Reliability (life). Each section of the dual indicator shall be subjected to a reliability (life) test conducted in 1-hour cycles for a total of 2,000 hours. At the beginning of each cycle, the indicator shall be energized and the pointers rotated to approximately full scale and shall remain energized in this position. After 57 minutes of each cycle, the pointers shall be rotated to the zero position and the indicator shall be de-energized. During these periods, the lighting shall be cycled as specified in the "Life" paragraph of MIL-L-27160. At the conclusion of this test, the indicator shall pass the test specified in 4.5.5. The total unit operating hours shall be divided by the total number of random failures which shall demonstrate an MTBF of 2,000 hours. A random failure is any failure of the indicator to function within the performance limits of this specification due to factors other than the following:

a. Exceeding the specified scheduled overhaul or replacement interval.
b. Improper maintenance.
c. Improper use.

For purposes of the MTBF computation, failures due to wearout prior to expiration of the scheduled replacement or overhaul interval shall be counted as random failures.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activities within the military service or defense agency, or within the military service's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The EGU-18/A indicator covered by this specification is intended for use in conjunction with the TRU-80/A oil pressure transmitter to remotely indicate the oil pressure in aircraft systems which require integrally lighted instruments.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. First article testing, when required (see 3.1).
c. Packaging (see 5.1).

6.3 Subject term (key word) listing.

remote
synchros
transmitter

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Air Force - 99

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